HW 6

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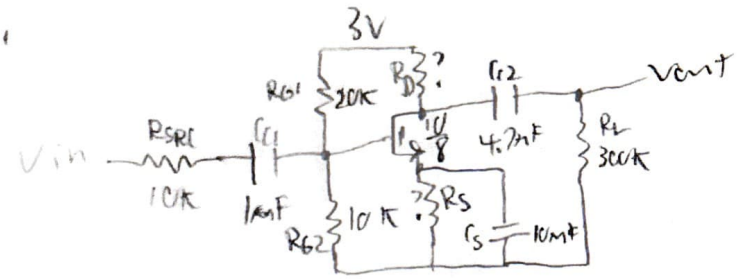
EEN 252

1.



$k_n' = 160\,mA/V^2$, $\lambda_n = .05$,

$V_T = .6V$, $C_{gs} = 50fF$

$C_{gd} = 15fF$, $C_{db} = C_{sb} = 5fF$

a. $V_G = 3V\left(\frac{10}{10+20}\right) = 1V$

$4\times10^{-6} = \frac{1}{2}\cdot 160\times10^{-6}\cdot\frac{10}{8}\left(1 - .6 - (4\times10^{-6}\cdot R_S)\right)^2$

$.04 = (.4 - R_S(4\times10^{-6})^2$ $V_{OV} = .2V$

$\boxed{R_S = 50k\Omega}$

b. $g_m = \frac{2I_D}{V_{OV}} = \frac{8\times10^{-6}}{.2} = 4\times10^{-5}\,S$

$= \frac{1}{25000k\Omega}$

$20k \| 10k = 6.6k\Omega$

$-3V/V = \frac{6.6k}{10k+6.6k}\left(-4\times10^{-5}\cdot\frac{300k(?)}{300k+?}\right)$

$-.0075 = -4\times10^{-5}\cdot\frac{300k(?)}{300k+?}$

$187.5 = \frac{300k(?)}{300k+?}$

$6.25\times10^{-4} = \frac{?}{300k+?}$

$? = 187.617k\Omega$

$\boxed{R_D = 187.617k\Omega}$

c. $\frac{3 - V_D}{187.617k\Omega} = 4\times10^{-6}$ $V_D = 2.2445V$

$\frac{V_S - 0}{50000} = 4\times10^{-6}$ $V_S = .2V$

$V_{DS} = V_D - V_S = 2.0445V$

$V_{OV} = V_{GT} = V_G - V_T - V_S = 1 - .6 - .2$

$= .2V$

$\boxed{.2 < 2.0445 \text{ so we in sat}}$

d. $\tau_{C1} = 1mF(10k + 6.6k) = .0167s \rightarrow 60$ rad/s ← dominant (highest freq)

$\tau_{C2} = 4.7mF(300k + 187.617k) = 2.2918s \rightarrow .4363$ rad/s

$\tau_{CS} = 10mF(50k \| 25k) = .167s \rightarrow 6$ rad/s

$\omega_L = \sum \frac{1}{\tau_{ii}} = 60 + 6 + .4363$

$= 66.4363$ rad/s

$\boxed{f_L = \frac{\omega_L}{2\pi} = 10.5737\,Hz}$

$C_{gs} = 50$ $C_{gd} = 5$ $C_{sb} = C_{db} = 5$ $g_m = 4 \times 10^{-5}$ $R_{in} = R_{sig} \| R_{G1} \| R_{G2}$ $R_{out} = R_L \| R_D \| r_o$

e.

$r_o = \frac{1}{4 \times 10^{-6}(.05)} = 5M$

$\tau_{in} = R_{in}(C_{gs} + C_{gd}(1 + g_m R_{out}))$

$= 4000 = R_{in}$

$\tau_{out} = R_{out}(C_{gd}(1 + \frac{1}{g_m R_{out}}) + C_{db})$ $= 112824.3201 = R_{out}$

$g_m R_{out} = 4.513$

dominant pole $\rightarrow$ $\tau_{in} = 5.3078 \times 10^{-10} s$ $f_H = \frac{1}{2\pi(\tau_{in} + \tau_{out})} = 50329399.43 \text{ Hz}$

$\tau_{out} = 2.6315 \times 10^{-9} s$

f.

Bandwidth is

10.5737 Hz to 50329399.43 Hz

total gap is

50329388.86 Hz